SC3102 Part 2 Cheat Sheet

1. Toolkit

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}\ (|q|<1) \qquad \sum_{n=1}^{\infty} q^n = \frac{q}{1-q}$$

$$\sum_{n=0}^{N-1} q^n = \frac{1-q^N}{1-q}\ (q\neq 1)$$

- **Geometric: first term/(1 − ratio)**, |ratio| < 1. $\sum_{a..b} q^n = (q^a - q^{b+1})/(1-q)$. $\sum_{n\geq 0}(az^{-1})^n = z/(z-a)$ if $|z| > |a|$; $\sum_{n\geq 1}(z/a)^n = (z/a)/(1-z/a)$ if $|z| < |a|$; $z = e^{j\Omega} \rightarrow$ DTFT.
- **Euler:** $\cos\vartheta = (e^{j\vartheta} + e^{-j\vartheta})/2$, $\sin\vartheta = (e^{j\vartheta} - e^{-j\vartheta})/(2j)$. $1/j = -j$, $e^{j\pi/2} = j$, $e^{j\pi} = -1$.
- **Sampling:** $x[n] = x_c(nT_s)$, $T_s = 1/f_s$. $\Omega = \omega T_s = 2\pi f/f_s$ (rad/sample).
- **Period of $\cos(\Omega_0 n)$:** write $\Omega_0/2\pi = m/N$ in lowest terms $\rightarrow N_0 = N$ (the denominator).
  - $\sin(\pi n/12)$: $1/24 \rightarrow N_0 = 24$. $\cos(3.5\pi n)$: $7/4 \rightarrow N_0 = 4$.
  - $\Omega_0/2\pi$ irrational (e.g. $\cos(n)$) $\rightarrow$ not periodic.
- **Sum of periodic signals:** $N_0 = \text{LCM}(N_1, N_2)$.
- **Nonzero periodic signal is never absolutely summable** ($\sum|x|$ adds the same block forever).
- **Aliasing:** Ω, $\Omega + 2\pi k$ and (for cos) $-\Omega$ give identical samples. In Hz: f, $f + k\cdot f_s$, $f_s - f$ look the same.
  - e.g. $\cos(3.5\pi n) = \cos(-0.5\pi n) = \cos(0.5\pi n)$.
  - Unique only if $0 \leq \Omega \leq \pi$, i.e. $f < f_s/2$. Nyquist: $f_s > 2f_{max}$.

2. Systems and Convolution

Property	Means	Quick test / red flags
Linear	$T(ax+bz) = a\cdot Tx + b\cdot Tz$	Double the input: does the output double? Fails with x^2, $x[n]\cdot x[n-1]$, + constant
Time-inv.	input shifted by r $\rightarrow$ output shifted by r	Fails if n appears outside $x[\cdot]$: $n\cdot x[n]$, $x[2n]$, $x[-n]$
Causal	y[n] uses only x[n], x[n−1], …	Fails with any x[n+1] (future input)
BIBO stable	$\|x\| \leq M \rightarrow \|y\| \leq C$	Bound term by term: $\|x[n]+x[n-1]+x[n]x[n-1]\| \leq 2M + M^2$

LTI: $h = T(\delta)$; $y = x * h$, $(x * h)[n] = \sum_m x[m]\cdot h[n-m]$

- **LTI causal** $\Leftrightarrow$ h[n] = 0 for n < 0. **LTI stable** $\Leftrightarrow \sum|h[n]| < \infty$.
- **Convolution facts:** $x * \delta[n-k] = x[n-k]$. $x * u$ = running sum $\sum_{m\leq n} x[m]$. $u * u = (n+1)u[n]$ ($\sum_m u[m]u[n-m]$ counts m = 0…n $\rightarrow$ n+1).
- **Step $\rightarrow$ impulse response:** $s = T(u) \Rightarrow h[n] = s[n] - s[n-1]$ (because $\delta[n] = u[n] - u[n-1]$).
- **Length:** linear conv of L_x and L_h samples has $L_x + L_h - 1$ samples.
- **FIR:** no y on the right; h = the coefficients; all poles at z = 0; always stable.
- **IIR:** feedback (y[n−k] on the right); h never ends; stable only if all poles inside |z| = 1.

3. DFT and DTFS

$$\text{analysis } X[k] = \sum_{n=0}^{N-1} x[n]\, e^{-j2\pi kn/N}$$

$$\text{synthesis } x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]\, e^{+j2\pi kn/N}$$

- **Bin k** $\leftrightarrow \Omega_k = 2\pi k/N \leftrightarrow f_k = k\cdot f_s/N$.
- **N = 4:** X = (matrix)·x. Synthesis x = ¼·(same matrix with $j \leftrightarrow -j$)·X.

Analysis row	n=0	n=1	n=2	n=3	Synthesis row (×¼)
k=0	1	1	1	1	1, 1, 1, 1
k=1	1	−j	−1	j	1, j, −1, −j
k=2	1	−1	1	−1	1, −1, 1, −1
k=3	1	j	−1	−j	1, −j, −1, j

- **Quick values:** X[0] = $\sum$x[n]. x[0] = (1/N)$\sum$X[k]. N = 4: X[2] = $x_0 - x_1 + x_2 - x_3$.
- **Real x:** X[N−k] = X[k]* (N = 4: X[3] = X[1]*); X[0] and X[N/2] are real. Rows 1 and 3 are conjugates.
- **Unknowns in x and X (T02 Q1d style):** fill in X[3] = X[1]* first, then use one synthesis row for a known x value.
- **DTFS:** same formulas, valid for all n only if N is a period (multiple of N_0); otherwise the repeated block $\neq$ the signal (leakage).
- **N_0 vs N:** N_0 = signal's period (calculate). N = DFT size (choose). No leakage $\Leftrightarrow$ N is a multiple of N_0.
- **DTFS by Euler (no sum):** write x as a sum of $e^{j2\pi kn/N}$; then X[k] = N × (coefficient of $e^{j2\pi kn/N}$). Use $e^{-j2\pi n/N} = e^{j2\pi(N-1)n/N}$.
  - e.g. $\cos(2\pi n/8) = \tfrac{1}{2}e^{j2\pi n/8} + \tfrac{1}{2}e^{j2\pi\cdot 7n/8} \rightarrow X[1] = X[7] = 8 \times \tfrac{1}{2} = 4$.
  - For sin: $1/(2j) = \tfrac{1}{2}e^{-j\pi/2}$.

4a. Circular Convolution

- **Multiplying DFTs gives circular convolution:** $\text{IDFT}_N\{X[k]\cdot Y[k]\} = x \circledast_N y$, not the linear one.
- **Wrap rule:** C[n] = L[n] + L[n+N] (+ L[n+2N] …): fold the tail of the linear result L back onto its start.
  - e.g. L = [−2, 1, 3, −6, 2, 8], N = 4 $\rightarrow$ C = [−2+2, 1+8, 3, −6] = [0, 9, 3, −6].
- **Circular = linear:** zero-pad both to $N \geq L_x + L_y - 1$.

R. Question-type Recipes (T01–T02)

On the sheet already: 1 Period N_0 *T01 Q1* $\rightarrow$ §9 *Simplify first*, then §1 m/N rule · **2 Same samples** *T01 Q2* $\rightarrow$ §1 *Aliasing* · **3 Properties** *T01 Q3* $\rightarrow$ §2 table + §9 *Causal*, *Linearity* · **4 Step $\rightarrow$ h** *T01 Q4* $\rightarrow$ §2: h[n] = s[n] − s[n−1], then test h (finite-length h $\Rightarrow$ stable) · **5 4-pt DFT matrix** *T02 Q1a* $\rightarrow$ §3 table (entry (k,n) = $(-j)^{kn}$, kn mod 4) · **9 Linear vs circular** *T02 Q4* $\rightarrow$ §4a.

6 Missing x / X values *T02 Q1d*: X[N−k] = X[k]* fills X $\rightarrow$ one synthesis row at a known sample, e.g. 4v[2] = $V_0 - V_1 + V_2 - V_3$ $\rightarrow$ synthesise the rest.

7 DTFS without the sum *T02 Q2b*: (i) N_0 $\rightarrow$ (ii) Euler-split into $e^{j2\pi kn/N}$ terms (or sin $\rightarrow \cos(\vartheta - \pi/2)$ first) $\rightarrow$ (iii) rename negative k $\rightarrow$ N−k $\rightarrow$ (iv) X[k] = N × weight $\rightarrow$ (v) check at n = 0.

8 N-pt synthesis = x? *T02 Q2c*: only if N is a multiple of N_0; else exact on 0…N−1 only (check x[N] vs x[0]).

10 DTFT of an exponential *T02 Q3*: (i) exists $\Leftrightarrow \sum|x| < \infty \Leftrightarrow$ |ratio| < 1 ($a^{-n}u[n]$: |a| > 1) $\rightarrow$ (ii) $x[n]e^{-j\Omega n} = q^n$, $q = \text{ratio}\cdot e^{-j\Omega}$, |q| = |ratio| < 1 $\rightarrow$ (iii) $\sum q^n = 1/(1-q) \rightarrow$ (iv) $y = x * x \Rightarrow Y = X^2$.

4b. The DTFT

$$\text{analysis } X(\Omega) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\Omega n}$$

$$\text{synthesis } x[n] = \frac{1}{2\pi}\int_0^{2\pi} X(\Omega)\, e^{j\Omega n}\, d\Omega$$

- **Repeats every 2π** in Ω (so the synthesis $\int$ can be over any 2π window: $0 \rightarrow 2\pi$ or $-\pi \rightarrow \pi$). **Exists if $\sum|x[n]| < \infty$**; u[n], $2^n u[n]$ and periodic signals fail.
- **Pairs (|a| < 1):**
  - $a^n u[n] \leftrightarrow 1/(1 - a\cdot e^{-j\Omega})$. **Why:** $a^n e^{-j\Omega n} = q^n$ with $q = a\cdot e^{-j\Omega}$, $|q| = |a|\cdot|e^{-j\Omega}| = |a|$ ($|e^{j\vartheta}| = 1$ always) $\rightarrow \sum q^n = 1/(1-q)$ needs |a| < 1. T02 Q3: $a^{-n}u[n] \rightarrow q = a^{-1}e^{-j\Omega}$, |q| = 1/|a| < 1 $\Leftrightarrow$ |a| > 1.
  - $a^{-n}u[-n-1] \leftrightarrow a\cdot e^{j\Omega}/(1 - a\cdot e^{j\Omega})$
  - $(n+1)a^n u[n] \leftrightarrow 1/(1 - a\cdot e^{-j\Omega})^2$ (= x * x)
- **Shift:** $x[n-k] \leftrightarrow e^{-j\Omega k}\cdot X(\Omega)$ (size same, only phase changes).
- **Convolution $\rightarrow$ product:** $Y(\Omega) = H(\Omega)\cdot X(\Omega)$. $H(\Omega)$ = DTFT of h = H(z) at $z = e^{j\Omega}$.
- **One frequency in $\rightarrow$ same frequency out:** $e^{j\Omega_0 n} \rightarrow H(\Omega_0)\cdot e^{j\Omega_0 n}$. |H| = gain, ∠H = phase shift.

9. Q&A Notes (things I asked about)

- **Period rule:** $\Omega_0 N = 2\pi m$ (N samples, m full laps, both whole); smallest m making N whole. $\Omega_0/2\pi = f/f_s$ = cycles per sample; periodic $\Leftrightarrow$ rational. Lowest terms: bottom = N_0, top = cycles.
- **Simplify first:** the rule gives when the *walker* returns; it overcounts only if every sample is 0: $\sin(\pi n) = \sin(3\pi n) = 0$ has period 1 (rule says 2). A sign on Ω or $\pm 2\pi$ aliasing doesn't change N_0. $\cos(-\vartheta) = \cos\vartheta$; $\sin(-\vartheta) = -\sin\vartheta$.
- **$T_s \neq T_0 \neq N_0$:** $T_s = 1/f_s$ (sampler, s); $T_0 = 1/f$ (wave, s); N_0 (samples, whole number). $N_0\cdot T_s = m\cdot T_0$.
- **Causal, formally:** if x = z up to time N, then the outputs are equal up to N. In practice: every x[·] index ≤ n for every n. Traps: x[2n], x[−n]. n·x[n] is fine.
- **Linearity:** $T(a\cdot x_1 + b\cdot x_2) = a\cdot Tx_1 + b\cdot Tx_2$, where a, b scale *values* (x[an] is time scaling, a different thing). **Disprove** with 1 counterexample that switches on the suspect term: for x[n]x[n−1], use v = δ[n] + δ[n−1]: T(v)[1] = 3, T(2v)[1] = 8 ≠ 6. **Prove** with algebra: product terms leave $(a^2-a)pq + (b^2-b)rs + ab(ps + rq)$, via $a^2pq = a\cdot pq + (a^2-a)pq$.
- **Convolution means:** $x = \sum x[m]\delta[n-m]$; each kick sends out x[m]·h[n−m]; add the copies (LTI only). [2, 1] * [1, 2, 3] = [2, 5, 8, 3].
- **Running sum, h = u** (§2): u in $\rightarrow$ (n+1)u[n] out. Bounded in, unbounded out $\rightarrow$ unstable (pole at z = 1).
- **Signal or system?** Transform a signal $\rightarrow$ X: what it contains. Transform h $\rightarrow$ H: what the system does to each frequency. Y = H·X. ROC for a signal $\rightarrow$ which side, whether the DTFT exists; for a system $\rightarrow$ causal, stable.
- **DFT as matrices:** $\omega = e^{j2\pi/N}$ (N = 4: ω = j), G[n,k] = ω^{nk}; G^H = conjugate transpose (G symmetric $\rightarrow$ just $j \leftrightarrow -j$). $X = G^H x$ (analysis: row k · x = X[k]); $x = (1/N)\cdot G\cdot X$ (synthesis); works because $G\cdot G^H = N\cdot I$. **Why −j:** the opposite spin stops a matching walker ($j^n(-j)^n = 1$ each time, sums to N); any other frequency cancels to 0.

5. Z-transform and its Pairs

analysis $X(z) = \sum_{n=-\infty}^{\infty} x[n]\,z^{-n}$, $z = r\cdot e^{j\Omega}$

synthesis (any r in ROC) $x[n] = \frac{r^n}{2\pi}\int_0^{2\pi} X(re^{j\Omega})\,e^{j\Omega n}\,d\Omega$

- **Meaning:** $X(re^{j\Omega})$ = DTFT of the faded signal $r^{-n}\cdot x[n]$. $|z| = r$ = fading, $\angle z = \Omega$ = frequency. On $|z| = 1$: $X(e^{j\Omega})$ = DTFT.
- **Undo the fading:** if $W(\Omega) = X(re^{j\Omega})$ then $w[n] = r^{-n}x[n]$, so $x[n] = r^n\cdot w[n]$ (= the synthesis formula; in practice invert with §7).
- **Tags:** finite list → $X(z) = \Sigma\, x[n]z^{-n}$. Coefficient of z^{-k} = $x[k]$ (future); of z^{+k} = $x[-k]$ (past).
- **Same thing, two forms:** $z/(z-p) = 1/(1 - p\cdot z^{-1})$.
- **DTFT inside X(z):** the Ω axis from $-\pi$ to π wraps round the unit circle ($\Omega = 0 \to z = 1$, $\pi/2 \to j$, $\pm\pi \to -1$). DTFT = 1 dial (Ω); Z = 2 dials (r, Ω). Building blocks are spirals $z^n = r^n e^{j\Omega n}$; a pole p means the signal contains p^n. e.g. $2^n u[n]$ faded with $r = 4$ gives 0.5^n (works for any $r > 2$, which is the ROC).

Given in the paper (no need to copy): $p^n u[n]$, $-p^n u[-n-1]$, $\rho^n\cos/\sin(\Omega_0 n)u[n]$ pairs; properties (linear, delay, $x[-n]$, $a^n x$, $x*h$); unilateral and bilateral shift rules.

Extra pairs (not given)	X(z)	ROC
$\delta[n-k]$	z^{-k}	all z (not 0 if k>0, not ∞ if k<0)
$p^{n-1}u[n-1]$	$1/(z-p)$	$\|z\| > \|p\|$
$-p^{n-1}u[-n]$ (opt.)	$1/(z-p)$	$\|z\| < \|p\|$
$n\cdot p^n u[n]$	$pz/(z-p)^2$	$\|z\| > \|p\|$
$-n\cdot p^n u[-n-1]$ (opt.)	$pz/(z-p)^2$	$\|z\| < \|p\|$
$(n+1)\cdot p^n u[n]$	$z^2/(z-p)^2$	$\|z\| > \|p\|$

$u[n]$ and $-u[-n-1]$: the given pairs with $p = 1$. **Ramp via u ∗ u:** $(n+1)u[n] = u * u \leftrightarrow U(z)^2 = z^2/(z-1)^2$; $n\cdot u[n] = (n+1)u[n] - u[n] \leftrightarrow z^2/(z-1)^2 - z/(z-1) = z/(z-1)^2$, $|z| > 1$ (pole on $|z| = 1$ → no DTFT). **Special case $\Omega_0 = \pi/2$:** $\rho^n\cos(\pi n/2)u[n] \leftrightarrow z^2/(z^2+\rho^2)$; $\rho^n\sin(\pi n/2)u[n] \leftrightarrow \rho z/(z^2+\rho^2)$; both $|z| > \rho$.

6. Poles, Zeros and ROC Rules

- **Poles** = roots of the denominator (×); **zeros** = roots of the numerator (○). First make ONE fraction in +powers of z and cancel: zeros hide in sums $(z/(z-0.6) + z/(z-1.2) = 2z(z-0.9)/(\dots))$. LCCDE: x-terms → zeros, y-terms → poles.
- **ROC is a ring** $r_1 < |z| < r_2$ with edges at poles; never contains a pole.
- **Which ring:** right-sided → outside the outermost pole · left-sided → inside the innermost pole · two-sided → a ring between poles (can be empty: $2^{|n|}$ needs $|z| > 2$ and $|z| < \frac{1}{2}$ → no ROC) · finite length → all z except maybe 0 and ∞.
- **P poles at different radii** → P + 1 possible ROCs, each a different $x[n]$.
- **Which way each pole runs:** pole inside the ring ($|p| \le$ inner edge) → right-sided $p^n u[n]$; pole outside ($|p| \ge$ outer edge) → left-sided $-p^n u[-n-1]$.
- **Stable (DTFT exists)** ⇔ unit circle $|z| = 1$ is in the ROC. A pole on $|z| = 1$ (e.g. $z = 1$) → no DTFT.
- **Causal** ⇔ ROC is outside the outermost pole **and** includes ∞ (numerator degree ≤ denominator degree in z).
- **Causal and stable** ⇔ every pole strictly inside the unit circle.

7. Inverting X(z)

1. **Factor the denominator** → poles. Mark the ROC's edges.
2. **Numerator has a factor z:** expand $X(z)/z = C/(z-p) + D/(z-q) \to X(z) = C\cdot z/(z-p) + D\cdot z/(z-q) \to p^n$-type terms.
3. **No factor z:** expand $X(z) = C/(z-p) + D/(z-q) \to p^{n-1}u[n-1]$-type terms.
4. **Cover-up:** $C = [(z-p) \times$ the fraction being expanded$]$ at $z = p$.
5. **Choose each term's side from the ROC** (section 6), then write $x[n]$.
6. **Special denominators:** repeated $(z-p)^2 \to n\cdot p^n$ or $(n+1)p^n$ pairs · quadratic $z^2 - 2\rho z\cdot\cos\Omega_0 + \rho^2 \to$ cos/sin pairs (ρ^2 = constant term, then $\cos\Omega_0$ from the middle) · only z^2 appears → set $w = z^2$, do partial fractions in w.

> **Example:** $X(z) = z/((z-\frac{1}{2})(z-2))$, $|z| > 2$. $X/z = 1/((z-\frac{1}{2})(z-2))$: $C = 1/(\frac{1}{2}-2) = -\frac{2}{3}$, $D = 1/(2-\frac{1}{2}) = \frac{2}{3}$. Both poles inside the ring → right-sided: $x[n] = [-\frac{2}{3}(\frac{1}{2})^n + \frac{2}{3}\cdot 2^n]u[n]$.

Series expansion (first few samples, right-sided): long-divide in powers of z^{-1}; $x[n]$ = coefficient of z^{-n}. e.g. $(1 + z^{-1})/(1 - \frac{1}{2}z^{-1}) = 1 + 1.5z^{-1} + 0.75z^{-2} + 0.375z^{-3} + \dots \to x[0..3] = 1, 1.5, 0.75, 0.375$.

Checks: causal x: $x[0] = \lim_{z\to\infty} X(z)$. Plug in n = 0, 1 and compare.

7b. Partial Fractions: the 3 Cases

Expand **X(z)/z** (long-divide first if top degree ≥ bottom), then × z back.

Factor below	Write	Find the numbers
$(z-p)$, distinct	$A/(z-p)$	cover-up at $z = p$
$(z-p)^2$, repeated	$A/(z-p) + B/(z-p)^2$	B: cover-up $(z-p)^2$; then A: plug in $z = 0$
$z^2 + bz + c$ (complex)	$(Cz + D)/(z^2 + bz + c)$	cover-up the others; then compare z^2, const. coeffs

Quadratic → cos/sin pairs: $\rho^2 = c$, $-2\rho\cos\Omega_0 = b$. Split $z(Cz + D) = C\cdot z(z - \rho\cos\Omega_0) + (D + C\rho\cos\Omega_0)\cdot z \to x = \rho^n[C\cdot\cos(\Omega_0 n) + K\cdot\sin(\Omega_0 n)]u[n]$, $K = (D + C\rho\cos\Omega_0)/(\rho\sin\Omega_0)$. Repeated: $z/(z-p)^2 \leftrightarrow n\cdot p^{n-1}u[n]$ (= extra pair ÷ p).

> **Quadratic e.g.** $X/z = 1/((z-1)(z^2+1)) = A/(z-1) + (Cz+D)/(z^2+1)$. $A = 1/(1+1) = \frac{1}{2}$. $1 = \frac{1}{2}(z^2+1) + (Cz+D)(z-1)$: z^2: $0 = \frac{1}{2} + C \to C = -\frac{1}{2}$; const: $1 = \frac{1}{2} - D \to D = -\frac{1}{2}$. $z^2 + 1$: $\rho = 1$, $\Omega_0 = \pi/2$, $K = (-\frac{1}{2} + 0)/1 = -\frac{1}{2}$. $|z| > 1$: $x = [\frac{1}{2} - \frac{1}{2}\cos(\pi n/2) - \frac{1}{2}\sin(\pi n/2)]u[n] = 0, 0, 1, 1, 0, \dots$ ✓

> **Repeated e.g.** $X/z = 1/((z-1)^2(z-2)) = A/(z-1) + B/(z-1)^2 + C/(z-2)$. $C = 1/1^2 = 1$; $B = 1/(1-2) = -1$; $z = 0$: $-\frac{1}{2} = -A - 1 - \frac{1}{2} \to A = -1$. $|z| > 2$: $x = [-1 - n + 2^n]u[n] = 0, 0, 1, 4, \dots$ ✓

8. Difference Equations (LCCDEs)

No initial conditions (bilateral, LTI) → H(z)

1. Transform every term: $y[n-k] \to z^{-k}Y$, $x[n-k] \to z^{-k}X$.
2. $H(z) = Y/X = (b_0 + b_1 z^{-1} + \dots)/(1 + a_1 z^{-1} + \dots)$. Multiply top and bottom by the highest power of z to read the poles.
3. One H(z), several ROCs → several systems; use section 6 for causal/stable. Leftover positive power of z on top (from $x[n+1]$) → not causal for any ROC.
4. $H = z^{-k}\cdot G \Rightarrow h[n] = g[n-k]$: invert G, then shift.

With initial conditions (unilateral, n ≥ 0)

unilateral $Y_+(z) = \sum_{n=0}^{\infty} y[n]\,z^{-n}$

Shift rules $\mathcal{Z}_+\{y[n-1]\}$, $\mathcal{Z}_+\{y[n-2]\}$ are given in the paper. An input starting at n = 0 has X_+ = its usual right-sided pair.

Recipe

1. Transform both sides with the shift rules.
2. Collect: $Y_+\cdot(1 + a_1 z^{-1} + \dots) = X_+$ − (initial-condition terms).
3. **Zero-input:** initial-condition terms ÷ $(1 + a_1 z^{-1} + \dots)$. **Zero-state:** $H\cdot X_+$.
4. Partial fractions (w = z^2 if only z^2 appears), invert with right-sided pairs (n ≥ 0).
5. **Complete:** $y = y_{zi} + y_{zs}$.
6. **Transient** = terms that die out (system poles inside $|z| = 1$). **Steady-state** = terms that last (input poles on $|z| = 1$).
7. **Remove the transient:** set $y[n] = y_{ss}[n]$ for all n; plug into the equation at n = 0 and n = 1; solve for $y[-1]$, $y[-2]$.
8. **Check:** compute y[0], y[1] straight from the equation. Initial value: $y[0] = \lim_{z\to\infty} Y_+(z)$. *(optional) Final value: $y[\infty] = \lim_{z\to 1}(1 - z^{-1})Y_+(z)$, only if all poles inside the unit circle except at most one at z = 1.*

A system is LTI only if its initial conditions are zero.

9. Q&A Notes (continued)

- **Why poles are ROC edges:** pole p ⇔ signal has p^n (×p each step); $\Sigma(p/z)^n$ needs $|z| > |p|$ → edge $|z| = |p|$.
- **1st-order LCCDE:** pole = number before $y[n-1]$ ($y = 0.5y[n-1] + x \to z/(z-0.5)$).
- **Zero:** input z^n → output 0.
- **ROC** = adds up *with* weight z^{-n}; stable = with no weight ($|z| = 1$). A blowing-up h can still have an ROC (Lab 4 Region 0).
- **sin or cos route, same X[k]:** $\sin\vartheta = \cos(\vartheta - \pi/2)$, and $1/j = e^{-j\pi/2}$ does the same $-\pi/2$ turn. T02: $\sin(2\pi n/6 + \pi/3) = \cos(2\pi n/6 - \pi/6)$ → weights $\frac{1}{2}e^{\mp j\pi/6}$.

R. Question-type Recipes (T03–T04)

On the sheet already: 13 Does an ROC exist? *T03 Q1d* → §6 *Which ring* (right part $|z| > a$, left part $|z| < b$; none if $a \ge b$) · **14 Damped cos** *T03 Q3a* → §7 step 6 + given cos/sin pairs · **15 Inverse** *T03 Q3b–c* → §7 (partial fractions, series) · **17 LCCDE → H, ROCs, h** *T04 Q2* → §8 steps 1–4 + §6.

11 X given on circle r *T03 Q1b*: $W(\Omega) = X(re^{j\Omega})$ = DTFT of $w[n] = r^{-n}x[n]$ → invert W with a §4b pair → $x[n] = r^n\cdot w[n]$. e.g. $r = 3$, $W = 1/(1 - (j/2)e^{-j\Omega}) \to w = (j/2)^n u[n] \to x = (3j/2)^n u[n]$.

12 X(z) and ROC of a signal *T03 Q2*: split into pieces → given pairs (+ §5 extras) (right-sided $|z| > |p|$, left-sided $|z| < |p|$) → ROC = overlap of all pieces; finite list → tags (all z except maybe 0/∞) → DTFT exists ⇔ $|z| = 1$ in ROC.

16 P-Z plot, all ROCs, x for each *T03 Q4*: factor top (○ zeros) and bottom (× poles) → P poles at distinct radii → P+1 ROCs → invert each pole by side (§6) → causal = outside the outermost pole; stable = ring containing $|z| = 1$; a pole on $|z| = r$ → no ROC includes that radius.

18 LCCDE + ICs, sinusoid in *T04 Q1*: → §8 *Recipe*. No transient ⇔ $y[-1] = y_{ss}[-1]$, $y[-2] = y_{ss}[-2]$; e.g. $y_{ss} = 4\cos(\pi n/2) \to y[-1] = 0$, $y[-2] = -4$.